Component-Based System Engineering (CBSE) condensed syllabus

Ordered logically from foundational concepts to advanced management:

Phase 1: Architectural Foundations (Middleware & APIs)

- **Chapter 1: Introduction to Component-Based Development** – Core definitions, the shift from object-oriented programming, and the business case for CBD.
- **Chapter 2: Component Architecture and Middleware** – Distributed systems, communication infrastructures, and standard component models (e.g., .NET, Enterprise JavaBeans).
- **Chapter 3: Interfaces, Contracts, and APIs** – Designing robust boundaries, API versioning, and contract-based design principles.

Phase 2: The CBSE Development Process

- **Chapter 4: The CBSE Lifecycle Process** – Specific engineering lifecycles for component-based systems, including domain engineering vs. application engineering.
- **Chapter 5: Designing for Reuse and Composition** – Principles of building loosely coupled components and techniques for wiring them together.
- **Chapter 6: Adapters and Mismatch Resolution** – Strategies for handling incompatible components using wrappers, proxies, and adapter design patterns.

Phase 3: Component Testing

- **Chapter 7: Component-Level and Integration Testing** – Black-box testing methodologies, validating interfaces, and verifying integrated systems.
- **Chapter 8: Third-Party Component Certification** – Quality assurance practices for evaluating, trusting, and verifying external software assets.

Phase 4: Security and Maintenance

- **Chapter 9: Component Security, Maintenance, and Versioning** – Addressing vulnerability risks in third-party code, regression testing, and handling "dependency hell" during updates.

Phase 5: Deployment and Management

- **Chapter 10: Component Deployment, Orchestration, and Repositories** – Packaging, containerization, managing runtime environments, and indexing assets in organizational repositories.